

Paper tables preview

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Table 1: Summary metrics for chat (non-reasoning) models (study 1).

Question Set	Model Metrics	Claude Haiku 3	DeepSeek-V3	GPT-4o	Gemini-2.5-Flash	Llama-3.1-70B	Llama-3.1-8B
BoolQ	Accuracy	78.15	82.90	85.30	84.54	78.59	69.32
	Confidence	89.76	92.94	93.26	99.30	90.60	85.98
	ECE	0.12	0.10	0.08	0.15	0.12	0.17
	(%) Rounded	99.88	91.33	98.36	89.93	88.17	96.28
HaluEval	Accuracy	52.12	52.12	52.12	52.12	52.12	52.12
	Confidence	88.65	67.16	76.49	60.29	63.73	71.21
	ECE	0.37	0.15	0.24	0.11	0.14	0.20
	(%) Rounded	100.00	100.00	100.00	98.60	99.11	98.72
LSAT-AR	Accuracy	29.07	34.88	36.05	96.51	38.37	33.72
	Confidence	67.39	63.84	85.58	97.48	89.53	88.80
	ECE	0.42	0.32	0.50	0.04	0.52	0.57
	(%) Rounded	89.53	90.70	90.70	98.84	100.00	94.19
SAT-EN	Accuracy	88.44	94.80	93.06	98.84	91.91	83.82
	Confidence	67.51	72.86	79.39	88.53	85.82	85.52
	ECE	0.22	0.24	0.15	0.10	0.10	0.12
	(%) Rounded	98.84	100.00	96.53	98.84	99.42	94.80
SciQ	Accuracy	94.07	97.09	96.88	96.68	95.38	91.06
	Confidence	84.35	86.55	91.95	93.44	94.68	94.51
	ECE	0.11	0.11	0.05	0.04	0.04	0.05
	(%) Rounded	99.80	98.99	99.40	78.39	97.59	97.29

Table 2: Summary metrics for reasoning models (study 1).

Question Set	Model Metrics	Claude-Sonnet-3.7	Claude-Sonnet-4	DeepSeek-R1	GPT-o3	Gemini-2.5-Pro
BoolQ	Accuracy	83.14	84.30	84.50	86.06	84.90
	Confidence	95.57	91.97	95.61	82.59	99.35
	ECE	0.12	0.08	0.11	0.04	0.15
	(%) Rounded	46.06	96.20	68.08	50.38	73.95
HaluEval	Accuracy	52.12	52.12	52.12	52.12	52.12
	Confidence	59.91	95.68	61.07	53.84	60.14
	ECE	0.08	0.45	0.11	0.02	0.11
	(%) Rounded	99.94	100.00	99.27	55.08	100.00
LSAT-AR	Accuracy	41.86	45.35	95.35	96.51	96.51
	Confidence	80.93	71.40	99.72	81.52	97.99
	ECE	0.39	0.33	0.05	0.15	0.01
	(%) Rounded	95.35	94.19	96.51	72.09	37.21
SAT-EN	Accuracy	97.69	98.84	94.22	97.69	98.84
	Confidence	84.16	82.63	87.06	75.55	92.74
	ECE	0.14	0.17	0.10	0.22	0.07
	(%) Rounded	97.11	100.00	94.80	83.82	63.01
SciQ	Accuracy	97.29	96.88	97.59	98.09	97.59
	Confidence	92.49	91.72	93.41	82.65	95.44
	ECE	0.05	0.05	0.04	0.15	0.03
	(%) Rounded	73.37	99.60	66.23	77.19	29.75

Table 3: Study 1 calibration metrics by question set and model (from the preregistered R analysis pipeline).

Question Set	Metric	Claude-Haiku-3	Claude-Sonnet-3.7	Claude-Sonnet-4	DeepSeek-R1	DeepSeek-V3	GPT-4o	GPT-o3	Gemini-2.5-Flash	Gemini-2.5-Pro	Llama-70B	Llama-8B
BoolQ	N	2503	2503	2503	2503	2503	2503	2503	2503	2503	2503	2503
	ECE	0.118	0.124	0.080	0.111	0.102	0.080	0.044	0.148	0.145	0.124	0.171
	Accuracy	0.781	0.831	0.843	0.845	0.829	0.853	0.861	0.845	0.849	0.786	0.693
	Overconfidence	0.116	0.124	0.077	0.111	0.100	0.080	-0.035	0.148	0.145	0.120	0.167
HaluEval	N	1790	1790	1790	1790	1790	1790	1790	1790	1790	1790	1790
	ECE	0.366	0.078	0.449	0.120	0.151	0.245	0.022	0.126	0.127	0.171	0.207
	Accuracy	0.521	0.521	0.521	0.521	0.521	0.521	0.521	0.521	0.521	0.521	0.521
	Overconfidence	0.365	0.078	0.436	0.089	0.150	0.244	0.017	0.082	0.080	0.116	0.191
LSAT	N	86	86	86	86	86	86	86	86	86	86	86
	ECE	0.444	0.391	0.331	0.034	0.329	0.495	0.150	0.038	0.015	0.535	0.595
	Accuracy	0.291	0.419	0.453	0.965	0.349	0.360	0.965	0.965	0.965	0.384	0.337
	Overconfidence	0.383	0.391	0.261	0.032	0.290	0.495	-0.150	0.010	0.015	0.512	0.551
SAT	N	173	173	173	173	173	173	173	173	173	173	173
	ECE	0.221	0.135	0.165	0.081	0.242	0.148	0.221	0.103	0.069	0.096	0.115
	Accuracy	0.884	0.977	0.988	0.942	0.948	0.931	0.977	0.988	0.988	0.919	0.838
	Overconfidence	-0.209	-0.135	-0.162	-0.072	-0.219	-0.137	-0.221	-0.103	-0.061	-0.061	0.017
SciQ	N	995	995	995	995	995	995	995	995	995	995	995
	ECE	0.109	0.049	0.054	0.044	0.105	0.054	0.154	0.039	0.031	0.042	0.054
	Accuracy	0.941	0.973	0.969	0.977	0.971	0.969	0.981	0.967	0.976	0.954	0.911
	Overconfidence	-0.097	-0.048	-0.052	-0.043	-0.105	-0.049	-0.154	-0.032	-0.021	-0.007	0.035

Table 4: Summary of model calibration across domains. Accuracy denotes mean correctness (WGD) or mean true probability (LifeEval, MedEval). Overconfidence = Mean Confidence – Accuracy. β_1 is the OLS slope of overconfidence on difficulty percentile. ECE_{SPD} and ΔECE show the effect of subjective probability distribution prompting.

Domain	Model	Accuracy	Mean Conf.	Overconf.	β_1	ECE	ECE_{SPD}	ΔECE (%)
WGD	Claude Haiku 4.5	0.385	0.274	−0.111	+0.433 ^{***}	0.123	0.081	−34.0 [*]
	Gemini 2.5 Flash	0.421	0.638	+0.217	+0.556 ^{***}	0.217	0.072	−67.0 ^{***}
	Llama 4 Maverick	0.372	0.588	+0.216	+0.471 ^{***}	0.216	0.074	−66.0 ^{***}
	GPT-5.4 Mini	0.427	0.443	+0.016	+0.237 ^{***}	0.046	0.087	+89.0 ^{***}
LifeEval	Claude Haiku 4.5	0.623	0.660	+0.038	+0.754 ^{***}	0.048	0.163	+235.5 ^{***}
	Gemini 2.5 Flash	0.645	0.766	+0.120	+0.820 ^{***}	0.121	0.099	−18.1 ^{**}
	Llama 4 Maverick	0.634	0.819	+0.185	+0.813 ^{***}	0.209	0.172	−17.8 ^{***}
	GPT-5.4 Mini	0.630	0.597	−0.033	+0.472 ^{***}	0.056	0.060	+5.6
MedEval	Claude Haiku 4.5	0.145	0.823	+0.678	+0.120 ^{***}	0.678	0.360	−46.8 ^{***}
	Gemini 2.5 Flash	0.148	0.870	+0.722	+0.130 ^{***}	0.722	0.420	−41.9 ^{***}
	Llama 4 Maverick	0.148	0.786	+0.637	+0.115 ^{***}	0.637	0.361	−43.4 ^{***}
	GPT-5.4 Mini	0.148	0.914	+0.766	+0.158 ^{***}	0.766	0.463	−39.5 ^{***}

^{*} $p < 0.05$, ^{**} $p < 0.01$, ^{***} $p < 0.001$

Table 5: Sensitivity of LifeEval difficulty metric to $Y_{\max}$. ρ is the Spearman rank correlation of difficulty percentiles against the reference ($Y_{\max} = 120$). Mean β_1 is the average OLS slope of overconfidence on difficulty percentile across four models.

$Y_{\max}$	ρ	Mean β_1
101	0.9530	0.7742
102	0.9573	0.7730
103	0.9620	0.7714
104	0.9669	0.7692
105	0.9716	0.7666
106	0.9759	0.7637
107	0.9799	0.7605
108	0.9835	0.7572
109	0.9867	0.7538
110	0.9894	0.7502
111	0.9918	0.7466
112	0.9938	0.7429
113	0.9954	0.7393
114	0.9968	0.7356
115	0.9978	0.7320
116	0.9987	0.7285
117	0.9993	0.7249
118	0.9997	0.7214
119	0.9999	0.7180
120	1.0000	0.7146 ←
121	0.9999	0.7112
122	0.9997	0.7079
123	0.9994	0.7046
124	0.9990	0.7014
125	0.9984	0.6982
126	0.9978	0.6950
127	0.9971	0.6919
128	0.9963	0.6888
129	0.9954	0.6858
130	0.9944	0.6828
131	0.9934	0.6799
132	0.9924	0.6770
133	0.9913	0.6742
134	0.9901	0.6714
135	0.9890	0.6686
136	0.9877	0.6659
137	0.9864	0.6632
138	0.9852	0.6606
139	0.9838	0.6581
140	0.9825	0.6555